

Organisational Barriers to the Lean Management Approach in Healthcare

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Abstract

To identify the potential organisational barriers to the lean management in healthcare organisations. Seven electronic databases were systematically searched for published studies relating to the challenges in lean implementation in healthcare systems. In total 412 articles were reviewed to select 23 studies which met the inclusion criteria and quality appraisal criteria. Research papers were selected following a screening process using the PRISMA tool. McMaster university framework was employed to access the quality of these selected papers. This process was to ensure appropriateness of the literature selected for this study. A thematic analysis was conducted on the results extracted from the research papers reviewed. Thematic analysis of the selected 23 studies revealed potential organisational barriers for implementing lean in healthcare organisations. Factors such as poor knowledge and perception of lean, management practices, organisational structure, employee resistance and poor collaboration are responsible for failures in lean implementation. Multiple barriers accounted for the challenging application of lean in healthcare. Successful lean implementation requires overcoming the functional and professional silos in healthcare organisations to reach common understanding and right perception of lean in healthcare. Furthermore, physicians and other clinicians are very important stakeholders, therefore healthcare organisations should encourage their leadership and involvement in lean implementation initiatives to improve the outcome of lean in healthcare. Lean management has the potential to improve efficiency and effectiveness of healthcare services, but lean principles, terminology and methods need to be adapted to healthcare settings for best outcomes.

Plain Language Summary

Organisational Barriers to the Lean Management Approach in Healthcare

This systematic review aimed at understanding the potential organisational barriers to the implementation of lean practices in healthcare. Lean approach has been successful in other industries for more than fifty years, but the application of lean in healthcare domain is still novice with limited empirical research on its validity and effectiveness in healthcare. To maximise the potential for success of lean, it is highly essential to anticipate and realize the challenges to lean initiatives in healthcare settings. Multiple factors accounted for the challenging application of lean in healthcare. Nevertheless, these challenges can be overcome by a system wide approach in staff education and training to give a clear understanding of lean principles and its benefits to improve employee perception and acceptance of lean. Altering the lean language for the context of healthcare will allow better permeability into healthcare organisations. Furthermore, physicians and other clinicians are very important stakeholders, therefore healthcare organisations need to encourage their involvement and leadership in lean initiatives and implementation for improved outcomes. Breaking silos by interdepartmental collaboration will facilitate communication and improve efficacy of lean processes throughout organisation. All the above, along with, careful 'adaptation' of lean tools and methods to the healthcare context will improve successful implementation of lean in healthcare,

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Keywords

lean healthcare, healthcare management, clinicians' involvement, clinicians' resistance, lean implementation, quality improvement, service efficiency

Introduction

Healthcare organisations all over the globe are under pressure due to factors like ageing population, rising costs of care, more costly technology and increasing customer expectations for high quality services (Mcintosh et al., 2014). In order to meet this demand, healthcare managers are constantly focusing on methods to reduce operational costs while maintaining or enhancing the quality of service (Antony et al., 2019). Some of the leading health care institutions have therefore adopted a promising management style called 'Lean management' (Toussaint & Berry, 2013). Lean is a systematic approach with key focus on identifying value of a process and eliminating any step that does not add value in that process, so that ultimately every step adds value to the entire process (Aherne & Whelton, 2010). In the year 1950, post-World War II, the need to do more in less sparked the birth of Toyota Production system which is now commonly known as 'Lean Manufacturing'. The National Health Service (NHS) in Great Britain and the Joint Commission institute in the United States of America have also encouraged lean management as a strategy for providing value to their patients while using less resources (Mazzocato et al., 2014). Lean has gained popularity and has become particularly essential in services (Klein et al., 2022). There has been a rapid increase in lean implementation in the past decade by many healthcare organisations reaping benefits like cost reduction as well as quality improvements (Daultani et al., 2015).

Lean in healthcare is still in the early phase of development if compared with its implementation in other industries. Although there is consensus in lean literature for potential of lean healthcare with significant benefits, Proudlove in 2008 stated that there is a lot of undergrowth to clear at hospitals which requires a deeper appreciation of lean for more complex lean implementations (Brandao de Souza, 2009).

The success of lean in healthcare depends on the ability to adapt and not just adopt lean in particular context to achieve continual improvement (Mazzocato et al., 2014). To be clear, healthcare is not similar to manufacturing, therefore lean cannot be directly replicated and relocated to healthcare systems. Some literature does not support extensive use of lean in healthcare due to complexity in adapting lean principles in healthcare and advise against simply duplicating manufacturing-based processes in healthcare (Mcintosh et al., 2014).

According to DeSouza and Pidd, there can be successful translation of lean practices in healthcare domain provided the inherent implementation barriers are identified and worked upon, all the way through the implementation process (Parkhi, 2019). Lean is being implemented since almost 50 years in other industries, but its implementation in healthcare still remains dubious. Hence, this systematic review aims to understand the potential organisational barriers to the lean management in healthcare.

Methods

Different type of studies was located, appraised and synthesised in this systematic review for analysis and identification of organisational barriers to implementation of lean management in healthcare organisations. Research papers were selected following a screening process using the PRISMA tool (PRISMA, 2020).

Data Search Strategy

A thorough and systematic search of databases such as, 'Business source premiere', 'Emerald Management Extra', 'Emerald premier: Business, Management and Strategy collection', 'Emerald premiere: Health & Social care', 'ProQuest', 'Elsevier Science direct', 'CINAHL', 'EBSCOhost' were used to find journal articles and research papers related to barriers in lean implementation in health care sector. The keywords used were 'Barriers, Challenges, Obstacles, Difficulties, Problem, and Complications' for the outcome of research question. Similarly, Mesh terms for lean were, 'Lean, Lean management, Lean six sigma, Lean manufacturing, Lean production'. Finally, for the term 'healthcare organisations', substitutes were, 'Healthcare Institutes, Health Services, Health sector, Clinics, Healthcare, Hospitals, Healthcare providers'. Additionally, departments within healthcare were also used such, 'emergency department, pharmacy' etc. All these keywords were used to search the above-mentioned databases using Boolean operators such as 'OR, AND or NOT'. Articles were then screened based on their titles. Selected studies were then transferred to document managing application named, 'Refworks' for further screening and also to save the searched material.

Data Extraction

All the studies selected for this review were in English language from the past 19 years, including research

Table 1. Inclusion and Exclusion Criteria.

Inclusion criteria	Exclusion criteria
Publication dates: January 2005 to December 2024. Empirical studies done using semi-structured interviews, questionnaires with oral feedback, surveys, reflection seminars, live project study & Interpretive inquiry, Service review by researchers. Research study conducted at healthcare organisations like hospitals, tertiary care centres, ambulatory care delivery systems, clinical departments in primary care, emergency units, laboratories. Participants were healthcare professionals, health service managers, physicians, senior executives in management teams or head nurses in management roles. Study articles where lean management interventions were used.	Systematic reviews, editorials, commentaries and non-primary research. Articles in languages other than English. Research conducted in other industries like manufacturing, hospitality or food industry which commonly use lean management. Participants outside healthcare setting or managers in other industries. Other similar interventions like total quality management, value analysis, operations research in healthcare organisations.

published between 2005 and 2024 to comprise all recent and relevant literature. Research studies of all countries were selected. All the pertinent empirical studies with quantitative, qualitative and mixed method have been included for this narrative synthesis. Table 1 details the inclusion and exclusion criteria used for data abstraction.

Study selection process was undertaken to include all the study papers that met the predetermined inclusion criteria. This was done in two stages. The first step included sifting through titles and abstract of all articles, followed by systematic screening of articles meeting the inclusion criteria. In the second phase of article selection, all the selected papers were further shortlisted by reading the complete text.

Quality Appraisal and Data Analysis

For qualitative systematic reviews such as this, authenticity and trustworthiness are judged instead of validity

and reliability (Bettany-Saltikov & McSherry, 2016). To assess the quality of the selected study papers, the McMaster University Quality Assessment Framework was used. This framework is useful for both qualitative and quantitative studies. Very clear guidelines are provided on the way's quality appraisal is conducted for studies and also provides advice while answering the questions. The framework includes basic terminology and can therefore be used by even students or new researchers (Bettany-Saltikov & McSherry, 2016). This tool comprised of questions on study population, selected participants with consent, study design, confounding factors and data collection methods followed by analyses. The study papers were then rated as weak, moderate or strong. Only moderate to strong study papers have been included for this review (see Table 2).

In the process of data extraction, findings and information that answer the research question were highlighted in the selected papers. A narrative data extraction form was prepared to present findings such

Table 2. Report of McMaster Appraisal Ratings per Study.

Author	Purpose of study	Quality appraisal
Kahm and Ingelsson (2019)	The study investigates first-line healthcare managers' views on their role and the conditions that influence their ability to drive improvement work based on Lean.	Strong
Hihnala et al. (2018)	The paper discusses health service manager's experience of lean management in a special healthcare unit of a hospital in Finland.	Strong
Hastle et al. (2016)	The paper investigates three possible explanations: maturity, complexity and value concept and analyse their impact on lean application in a Danish hospital.	Strong
Stanton et al. (2014)	This study aims to analyse a process improvement project based on lean six sigma technique in emergency department of a large Australian hospital.	Moderate
D. Hung et al. (2017)	This study identifies contextual factors most critical to implementing Lean redesigns across primary care clinics in a large, ambulatory care delivery system in the United States of America.	Strong
Edwards et al. (2011)	This study is attempting to identify possible barriers to the successful implementation of lean management in healthcare.	Moderate

(continued)

Table 2. (continued)

Author	Purpose of study	Quality appraisal
Fine et al. (2009)	The study highlights how five Canadian hospitals began their Lean initiatives and discuss the lessons learned to help other organisations in applying Lean in a Canadian setting.	Strong
Kinder and Burgoyne (2013)	The study paper explores the reason for high failure rate in NHS lean rapid improvement project from the perspective of information processing.	Strong
Grove et al. (2010)	This study presents challenges identified during lean implementation in a health visiting service within a large primary care trust in NHS, UK.	Strong
Cheng et al. (2015)	The purpose of this study is to disseminate findings from a lean healthcare project to demonstrate relevance & extent that lean can be applied in NHS.	Moderate
DeSouza and Pidd (2011)	The study article analyses lean implementation barriers based on its experience of using lean methods in NHS trusts.	Strong
Azevedo et al. (2021)	This study identifies the facilitators and barriers to Lean Enterprise Transformation (LET) Programs in 10 Veterans Health Administration (VA) centres in the USA.	Moderate
Costa et al. (2015)	This study evaluates five sectors of two Brazilian hospitals have after implementing lean methods in their operations.	Strong
Aij et al. (2013)	The purpose of this study is to investigate the experience of leaders in implementation of lean after a lean training programme in a large teaching hospital in Netherland.	Strong
Saha et al. (2024)	This study investigates the role of leadership and cultural transformation in facilitating Lean Six Sigma (LSS) practices in clinical pharmacy settings to reduce medicine waste within the UK National Health Services (NHS).	Strong
Ruiz and Ortiz (2016)	The study purpose is to identify the barriers and enablers to lean approach in healthcare based on case studies in Columbian health sector.	Moderate
Escuder et al. (2018)	The study purpose is to outline barriers in lean implementation in Uruguayan health care.	Moderate
Fournier et al. (2021)	This study attempts to understand the physicians' behaviours in Canadian hospitals during Lean transformations and the factors that drive resistance. It also provides insight into how organisations can improve the engagement of medical staff in their Lean efforts to offset Resistance.	Strong
Mutingi et al. (2017)	The study investigates barriers and enablers for lean adoption and implementation in the context of medical laboratories in Namibia.	Strong
Leite et al. (2024)	This study attempts to understand the underlying impact of barriers in healthcare operations on lean implementation and sustainability in emergency care in the Brazilian public healthcare system.	Strong
Jorma et al. (2016)	The study evaluates outcome of lean management tools in a hospital in Finland.	Strong
D. Hung et al. (2015)	The purpose of this study is to highlight the key facilitators and barriers to Lean process at primary care providers based on a study in a large ambulatory care system in United States.	Strong
Halling and Wijk (2013)	This study compares similarity and differences in lean concepts and barriers to lean implementation in a large manufacturing company with two Emergency health care units in Sweden.	Strong

as, name of author, year of study, study design, data collection methods and the outcome or key findings. This data was categorised to reflect the perceived barriers in implementation of lean methods in healthcare settings. Table 3 is the data extraction form of this study, depicting the key findings and organisational barriers. Thematic analysis was used to identify key themes from the selected papers. The repeated reading of data extracted from the papers was carried out to identify recurrence themes related to organisational barriers in lean management in healthcare. From these themes final themes were developed iteratively to examine the phenomena studied (Ross, 2012).

Results

Total 23 study papers met all the criteria for this systematic review (see Figure 1). Study papers have been included from many countries: UK (5), Finland (2), Sweden (2), Denmark (2), USA (2), Namibia (1), Uruguay (1), Columbia (1), Netherland (1), Brazil (2), Canada (2) and Australia (1). Among the 23 papers, 14 papers had qualitative study design, 6 were mixed method studies and remaining 3 were quantitative study papers. The study population included, chief executive officers, healthcare managers of all cadre, physicians, nurses and employees from various healthcare settings ranging from large

Table 3. Data Extraction Form of the Study.

Author	Purpose of study	Study population	Study design	Quality appraisal	Study method	Key findings
Kahm and Ingelsson (2019)	The study investigates first-line healthcare managers' views on their role and the conditions that influence their ability to drive improvement work based on Lean.	First line managers in Swedish healthcare organisation	Quantitative	Strong	Questionnaire with oral feedback	Employees do not involve and engage in lean change initiatives. There is no cooperation and agreement between different professionals. Lacking executive support in improvement work based on lean. Forced change by management causes lack of ownership and most quality initiatives fail because staff has no time.
Hihnala et al. (2018)	The paper discusses health service manager's experience of lean management in a special healthcare unit of a hospital in Finland.	Seven nurse managers and seven medical managers	Qualitative	Strong	Thematic interviews	Managers experience lack of time in lean implementation due to increased documentation & reporting. Physical structure also limits development of day-to-day functions. Employees lack motivation for lean methods.
Hastle et al. (2016)	The paper investigates three possible explanations: maturity, complexity and value concept and analyse their impact on lean application in a Danish hospital.	CEO & two consultants in a Danish university hospital	Qualitative	Strong	Interviews	There is limited practical experience of lean process in hospitals. A scattered picture and short life span of lean methods in hospitals also hampers research and development. The complexity of healthcare organisations makes lean difficult, also hospital context makes it difficult to define value concepts.
Stanton et al. (2014)	This study aims to analyse a process improvement project based on lean six sigma technique in emergency department of a large Australian hospital.	CEO, senior executives, doctors, nurse managers & quality improvement staff of tertiary care hospital in Australia	Qualitative	Moderate	Semi structured interviews	There is no strong evidence that process redesign /lean is effective. Staff has limited understanding of lean and changes in other areas of hospital. They lack clarity of concepts about 'customer' and 'value for customer'. Occupational demarcations complicate and constrain efforts of process improvisation. Also, Organisational politics & hierarchical structure influences resources and nature of work.

(continued)

Table 3. (continued)

Author	Purpose of study	Study population	Study design	Quality appraisal	Study method	Key findings
D. Hung et al. (2017)	This study identifies contextual factors most critical to implementing Lean redesigns across primary care clinics in a large, ambulatory care delivery system in the United States of America.	Sixty-one frontline physicians, 21 physician leaders, 23 operational leaders at 10 clinics in a large ambulatory care delivery system	Qualitative	Strong	In-depth interviews	There is low staff engagement level & intensity of lean implementation process. Managers lack good leadership and resist top-down approach in lean implementation. A failure in documenting improvement in efficiency by results in lack of evidence and cause scepticism and unwillingness to embrace lean. Physician are resistant who consider lean management-based interventions as a potential threat to their authority. Difficulty was reported in transferring lean concept from industrial context to healthcare. Medical professionals had difficulty absorbing the tools, ideas and principles of lean. Some managers said lean cannot be applied to all processes and activities in healthcare organizations.
Edwards et al. (2011)	This study is attempting to identify possible barriers to the successful implementation of lean management in healthcare.	Three hospital wards & one home care nursing unit data in Denmark	Qualitative and quantitative	Moderate	Reflection seminar, Interview & Questionnaire	Negative connotations used in lean language were said to evoke negative image for ruthless efficiency. Even Staff had myths about lean initiatives, 'Lean means job cuts'. There is very less physician engagement. Leaders find it challenging to sustain lean processes.
Fine et al. (2009)	The study highlights how five Canadian hospitals began their Lean initiatives and discuss the lessons learned to help other organisations in applying Lean in a Canadian setting. The study paper explores the reason for high failure rate in NHS lean rapid improvement project from the perspective of information processing. This study presents challenges identified during lean implementation in a health visiting service within a large primary care trust in NHS, UK.	CEO, director of quality and patient safety, senior project manager & lean co-ordinator in (5) Canadian Hospitals	Qualitative	Strong	Semi-structured interviews	Team leaders found that information processing system and the decision hierarchy make sustainable success difficult for lean. Functional silos were found to exist due to different professional groups. Process variability. Managers highlight poor knowledge of basic lean philosophy by project team, with difficulty in identifying 'waste' and 'customer' in healthcare. Also, there was limited communication & leadership in hospitals. Leaders only focussed on achieving targets.
Kinder and Burgoyne (2013)		Two team leaders and three project participants in a Scottish health board	Qualitative	Strong	Interviews, live project study & Interpretive inquiry	
Grove et al. (2010)		Service Manager, researcher & external lean consultants in an NHS trust. (n = 14)	Qualitative	Strong	Semi-structured interviews	

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Table 3. (continued)

Author	Purpose of study	Study population	Study design	Quality appraisal	Study method	Key findings
Cheng et al. (2015)	The purpose of this study is to disseminate findings from a lean healthcare project to demonstrate relevance & extent that lean can be applied in NHS.	Secondary mental health NHS trust in England	Qualitative & Quantitative	Moderate	Service review by researcher for 1 year included Interviews, questionnaire, surveys & secondary data study	Service managers found poor quality of referral information flow between departments. Also, the staff had difficulty in defining customer.
DeSouza and Pidd (2011)	The study article analyses lean implementation barriers based on its experience of using lean methods in NHS trusts.	Director, managers and physicians in three acute care NHS trusts.	Quantitative	Strong	Interviews	Lean terminology from manufacturing influenced perception of lean by healthcare staff. Most employees lacked professional skills doing faulty data collection and measurement. Multiple professional groups led to formation of functional and professional silo. Hierarchy & management role also were responsible for employee resistance. Organisations lacked momentum for continuous improvement
Azevedo et al. (2021)	This study identifies the facilitators and barriers to Lean Enterprise Transformation (LET) Programs in 10 Veterans Health Administration (VA) centres in the USA.	268 key stakeholders which involve medical centre directors, other executive leadership, systems redesign/process improvement directors, value stream process owners (those tasked with completing a value stream, such as the chief of the emergency department), middle managers (e.g. physician manager, social work manager), frontline supervisors (e.g. emergency department nursing supervisor, medical assistant supervisor), frontline staff (e.g. physicians, nurses, clerks) and Lean coaches.	Mixed methods	Moderate	Three rounds of interviews, one in-person site visit and two follow-up telephone visits, at 6-month intervals. The telephone interviews were used to follow the trajectory of implementation at each site. 227 semi structured interview sessions (including focus groups) representing 268 key stakeholders	Merely using Lean tools was not sufficient as there were no on-going opportunities for Lean education to solidifying knowledge of Lean concepts and management strategies. The training on lean was not adequate to get the buy-in at the levels that are necessary to really make Lean work effectively.

(continued)

Table 3. (continued)

Author	Purpose of study	Study population	Study design	Quality appraisal	Study method	Key findings
Costa et al. (2015)	This study evaluates five sectors of two Brazilian hospitals have after implementing lean methods in their operations.	Five sectors in two hospitals of Brazil	Qualitative & Quantitative	Strong	Multiple case study by interviews, observation & archival source	Hierarchical organisational structure leads to poor management relationship with staff. Specialised consultants don't support lean methods due to lack of knowledge about lean. Outsourcing any hospital sector makes lean implementation complicated. Continuity & sustenance of lean is a challenge for management. Lack of management support and commitment and poor allocation of resources. There is no strategy and purpose of lean in most organisations. Multi-disciplinary noncollaboration and organisational silos in healthcare makes lean implementation difficult. Employees lack education and training on lean management, hence resistant to change.
Aij et al. (2013)	The purpose of this study is to investigate the experience of leaders in implementation of lean after a lean training programme in a large teaching hospital in Netherland.	Thirty-one medical, surgical and nursing professionals with supervisory experience	Qualitative	Strong	Semi-structures and in-depth interviews	The absence of leadership support in integrating waste reduction practices demotivated pharmacists in LSS process. Hierarchical structures created a disconnect between junior pharmacists and the leadership team although junior pharmacists being ideally positioned to identify and address waste-related issues because they heavily involved in clinical services.
Saha et al. (2024)	This study investigates the role of leadership and cultural transformation in facilitating Lean Six Sigma (LSS) practices in clinical pharmacy settings to reduce medicine waste within the UK National Health Services (NHS).	Four Junior Pharmacists, five Senior Pharmacists, four Lead Pharmacists and one Chief pharmacist in the NHS	Qualitative	Strong	A qualitative case study employing semi structured interviews	Most centres had no lean training for employees. Employees lacked engagement & understanding of lean. There was clear lack of commitment by management. No time and resources for improvement were allotted. An organisational culture of lean is missing.
Ruiz and Ortiz (2016)	The study purpose is to identify the barriers and enablers to lean management in healthcare based on case studies in Colombian health sector.	Ten health care centre in Columbia	Qualitative	Moderate	Semi-structured interviews	

(continued)

Table 3. (continued)

Author	Purpose of study	Study population	Study design	Quality appraisal	Study method	Key findings
Escuder et al. (2018)	The study purpose is to outline barriers in lean implementation in Uruguayan health care.	850 Managers	Quantitative	Moderate	Online survey	Lean terminology is misunderstood. There is no skilled facilitator or communicator for lean improvement events. Employees lack training, commitment & support of top managers. Stakeholder participation is required with good managerial skills. Key performance indicators are needed to motivate employees. Hierarchical structure and Organisational silo exist that inhibit lean implementation. Culture of improvement is needed to overcome resistance to lean in healthcare.
Fournier et al. (2021)	This study attempts to understand the physicians' behaviours in Canadian hospitals during Lean transformations and the factors that drive resistance. It also provides insight into how organisations can improve the engagement of medical staff in their Lean efforts to offset resistance.	Chief Executive Officer (CEO), the Chief Human Resources Officer (CHRO), the Chief Medical Officer (CMO), the Director of the Lean program, a middle manager and three front-line physicians who had participated in Lean change initiatives	Longitudinal qualitative case studies in Canadian hospitals	Strong	Ninety-nine interviews conducted with various stakeholders using open-ended questions related to the implementation of Lean in their organisations	Lean clashed with the dominant medical professional practice found in healthcare organisations, resulted in physicians displayed resistance behaviour towards lean implementation. The most resistance behaviours were triggered when physicians perceived that a proposed Lean change would target their medical practices. Resistance behaviours were also triggered by perceptions of efficiency-driven Lean implementation which was based on industrial process. Lack of support by management and financial constraints reported by laboratory services. Employees lack of conceptual knowledge about lean leads to staff resistance. Absence of lean culture in laboratory.
Mutingi et al. (2017)	The study investigates barriers and enablers for lean adoption and implementation in the context of medical laboratories in Namibia.	Seventy-two medical laboratory services in Namibia	Qualitative & Quantitative	Strong	Questionnaire	Professional boundaries and traditional hierarchical medical dominance reduced the involvement of physicians in the lean process in healthcare. Healthcare professionals have lack of acceptance for Lean management because it was based on industrial processes.
Leite et al. (2024)	This study attempts to understand the underlying impact of barriers in healthcare operations on lean implementation and sustainability in emergency care in the Brazilian public healthcare system.	Nurses, physicians and coordinators who have been implementing the lean model in the research sites	Qualitative	Strong	Exploratory illustrative case studies within three emergency departments employing 32 semi-structured interviews were conducted with nurses, physicians and coordinators	

(continued)

Table 3. (continued)

Author	Purpose of study	Study population	Study design	Quality appraisal	Study method	Key findings
Jorma et al. (2016)	The study evaluates outcome of lean management tools in a hospital in Finland.	110 Healthcare professionals from 15 hospitals in Finland	Quantitative & Qualitative	Strong	Questionnaire & webropol survey	Employee resistance to lean implementation due to lack of information about lean. Discrepancy in professional groups.
D. Hung et al. (2015)	The purpose of this study is to highlight the key facilitators and barriers to Lean process at primary care providers based on a study in a large ambulatory care system in United States.	Forty-five physicians from three clinical departments in primary care setting	Qualitative	Strong	In depth interviews & focus groups	Leaders agree that management driven change is needed. There is inadequate data collection and lack of visibility of the progress achieved. Physicians resist standardisation of work. There has been difficulty transferring responsibility to others as lean flow managers. The staff Lacks time for training due to cross coverage of patient care during lean improvement events.
Halling and Wijk (2013)	This study compares similarity and differences in lean concepts and barriers to lean implementation in a large manufacturing company with two Emergency health care units in Sweden.	Team leaders of two Emergency unit in Sweden	Qualitative	Strong	Semi-structured interviews	Leaders found that there is limited knowledge of lean in consultants. Employees lack vision and there is poor leadership. Insufficient co-operation and communication in staff. Organisational silo exist which acts as barrier for flow of services. Since improvement projects are time consuming there is reactive culture towards lean.

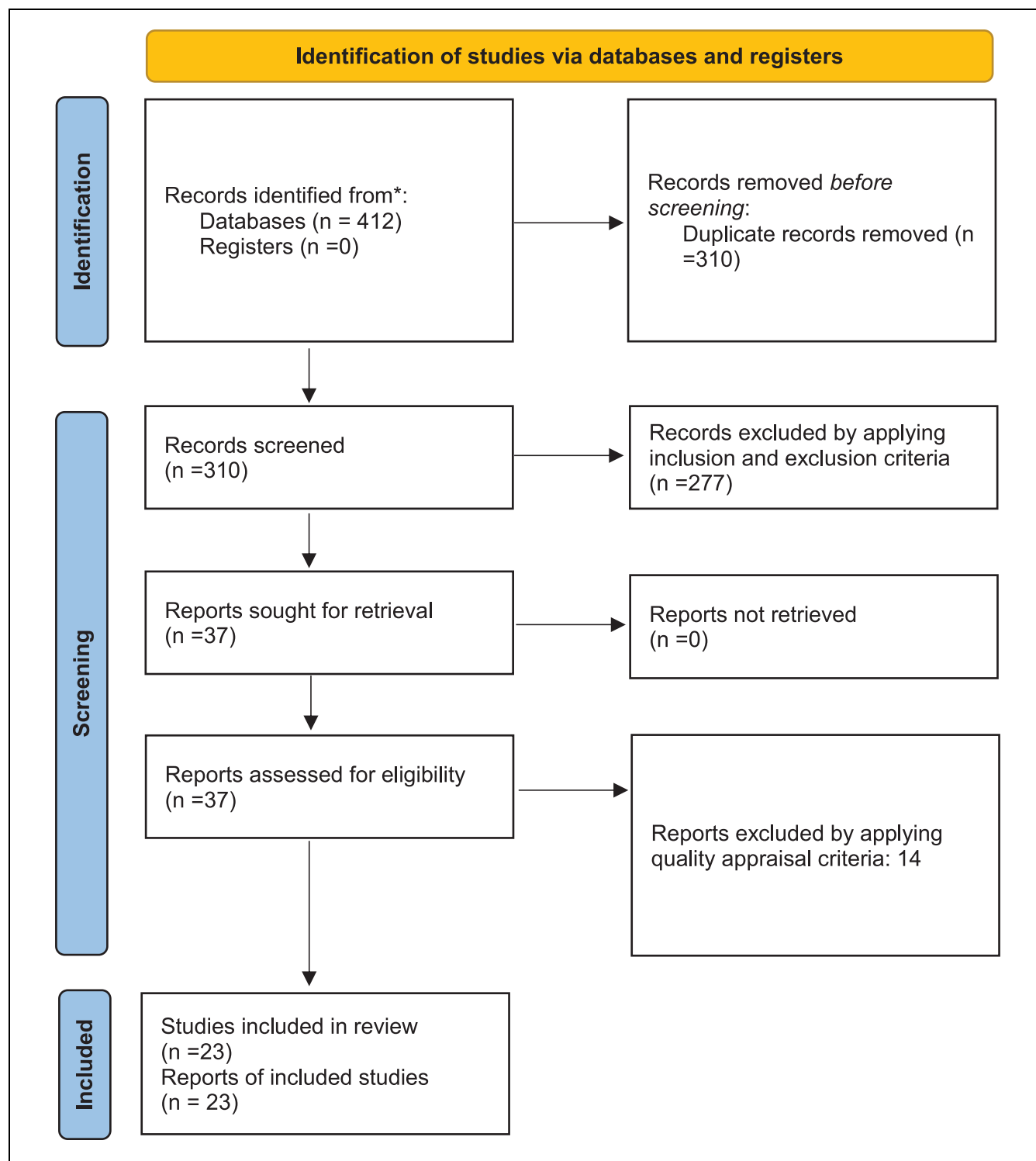


Figure 1. PRISMA flow diagram.

Source. Adapted from Page et al. (2021).

tertiary care hospitals to primary care facilities, emergency departments, clinics, laboratory services etc. All the article findings were then used to create initial

common themes which emerged throughout the key findings. Tables 4 to 7 depict thematic analysis of organisational barriers in implementing lean methods.

Table 4. Theme 1: Poor Perceptions and Knowledge of Lean.

Authors	Initial theme	Final theme
Aij et al. (2013), Azevedo et al. (2021), Cheng et al. (2015), Costa et al. (2015), DeSouza and Pidd (2011), Edwards et al. (2011), Escuder et al. (2018), Fine et al. (2009), Grove et al. (2010), Halling and Wijk (2013), Hastle et al. (2016), Mutingi et al. (2017), Stanton et al. (2014)	Employees, managers and consultants lack conceptual knowledge about lean. Lean terminology from manufacturing background is misunderstood. Staff has myths about lean process due to negative connotations used in lean language. There is limited evidence of lean effectiveness and practical experience of lean. Hospitals lack trained & skilled professionals for lean implementation	Poor perception and knowledge of lean

Table 5. Theme 2: Management Practice.

Authors	Initial theme	Final theme
Aij et al. (2013), Costa et al. (2015), DeSouza and Pidd (2011), Escuder et al. (2018), Fine et al. (2009), Grove et al. (2010), Halling and Wijk (2013), Hihnala et al. (2018), D. Hung et al. (2015, 2017), Kahm and Ingelsson (2019), Mutingi et al. (2017), Ruiz and Ortiz (2016), Saha et al. (2024)	A strategy and purpose of lean is missing. Management support for initiatives and resource allocation is not available. Managers lack necessary skills and leadership qualities necessary for lean implementation. Visibility of progress is missing due to inadequate data collection. Lack time for employee training & process improvements. Managers not successful at maintaining continuity and sustaining lean initiatives.	Management practices

Table 6. Theme 3: Organisational Structure.

Authors	Initial theme	Final theme
Aij et al. (2013), Costa et al. (2015), Edwards et al. (2011), Escuder et al. (2018), Grove et al. (2010), Halling and Wijk (2013), Hastle et al. (2016), Hihnala et al. (2018), Jorma et al. (2016), Kahm and Ingelsson (2019), Kinder and Burgoyne (2013), Leite et al. (2024), Saha et al. (2024), Stanton et al. (2014)	Occupational demarcations complicate & constrain process of improvisation. Difficulty transferring lean concept from industrial context to healthcare. Complexity of healthcare organisations and process variability makes lean implementation difficult in healthcare. Physical structure also limit day to day development functions. Discrepancy in multiple professional groups. Information processing system & decision hierarchy make sustainable success difficult. Functional & professional Silo due to multiple skilled groups act as flow barrier.	Organisational structure

Hence, final thematic analysis of barriers to lean in healthcare gave us four final themes as below:

- 1- Poor perception and knowledge of lean
- 2- Management practices
- 3- Organisational structure
- 4- Employee resistance and poor collaboration.

Major Barriers in Lean Implementation in Healthcare Organisations

Poor Perception and Knowledge of Lean. Lack of knowledge and understanding of lean acts as a prominent barrier with 10 studies indicating that employees do not have lean awareness in most healthcare organisations (Grove

Table 7. Theme 4: Employee Resistance and Poor Collaboration.

Authors	Initial theme	Final theme
Aij et al. (2013), Costa et al. (2015), DeSouza and Pidd (2011), Escuder et al. (2018), Fine et al. (2009), Fournier et al. (2021), Halling and Wijk (2013), Hihnala et al. (2018), D. Hung et al. (2015, 2017), Jorma et al. (2016), Kahm and Ingelsson (2019), Leite et al. (2024), Mutingi et al. (2017), Ruiz and Ortiz (2016)	Employees don't involve and engage in lean change initiatives. Physician resistance to process standardization. Employee resistance due to top-down approach. Lack of motivation in employees for lean processes. Inefficient communication & reactive culture towards lean. Absence of lean culture & organisational momentum necessary for sustaining lean implementation.	Employee resistance and poor collaboration

et al., 2010; Mutingi et al., 2017). ‘*Poor understanding of lean principles by the project team, were a barrier to success. The staff were introduced to the top 10 per cent of the “lean iceberg” and were asked to perform tasks*’ (Grove et al., 2010). Moreover, three studies argue that employees find it difficult to define and translate key concepts such as, ‘who the customer is?’ or ‘what creates value?’ for the customer (Cheng et al., 2015; Grove et al., 2010; Stanton et al., 2015). This lack of knowledge is also a result of unavailability of professional training to healthcare employees. Six studies have mentioned that no training was given for lean processes and that staff had limited practical experience of lean process in hospital (Aij et al., 2013; Azevedo et al., 2021; Costa et al., 2015; Escuder et al., 2018; D. Hung et al., 2015; Ruiz & Ortiz, 2016). Further to lack of knowledge, a poor perception of lean in healthcare is because lean language evokes negative image of sheding jobs. Also, the manufacturing background acts as a catalyst in reducing acceptance of lean in healthcare (DeSouza & Pidd, 2011; Fine et al., 2009; Grove et al., 2010). Failure in documenting efficiency or faulty data collection techniques are factors responsible for not having strong evidence to support efficacy of lean methods. Seven study papers find this to be a reason for the partial acceptance of lean by physicians and other healthcare employees (DeSouza & Pidd, 2011; Escuder et al., 2018; Hastle et al., 2016; D. Hung et al., 2015, 2017; Jorma et al., 2016; Stanton et al., 2014).

To address this issue management needs to emphasise that lean management in healthcare is about improving patient care not just a cost-cutting exercise. They should highlight the benefits for clinicians and other healthcare employees by clearly articulating key lean management concepts such as value, flow and waste reduction relating them to healthcare practice. Such attempts would enable clinicians to understand that lean can reduce their workload, improve clinical workflows, enhance their quality of work life and eventually lead to better patient

outcomes. At this juncture, developing training programmes for clinicians and other healthcare employees is an important organisational intervention. Such training can reduce the fears and concerns that stem from their perceptions regarding implementation of lean management in healthcare (see Abou Hashish & Abdel Aal, 2019; Hihnala et al., 2018; Rosa et al., 2021).

Management Practices. One of the most commonly mentioned factors in this category has been lack of support and commitment from top management. Seven studies have found that management support for lean initiatives is missing and that they lack commitment to lean processual changes (Aij et al., 2013; DeSouza & Pidd, 2011; Escuder et al., 2018; D. Hung et al., 2015; Mutingi et al., 2017; Ruiz & Ortiz, 2016; Saha et al., 2024). One of the participant interviewees from the reviewed studies offered the following justification, ‘*The problem is that top management sits in their ivory towers. They trust that everything will work out fine on the work floor. I think there is too much distance between management and their teams because they are always busy*’ (Aij et al., 2013). According to Kahm and Ingelsson (2019) healthcare domain needs a reflective managerial change where manager’s actions interact and communicate across their co-workers. However, seven studies described that manager lacked necessary skills and leadership style for successfully carrying out lean initiatives (Costa et al., 2015; DeSouza & Pidd, 2011; Escuder et al., 2018; Grove et al., 2010; Halling & Wijk, 2013; D. Hung et al., 2017; Ruiz & Ortiz, 2016). Other challenges created by the management accounted for lack of time for training and process improvements. Managers reported to have experienced shortage of time for lean process implementation as lean methods increased documentation and reporting of work, leaving them with less time for managerial activities (Hihnala et al., 2018; D. Hung et al., 2015). Further to that, four other studies found that insufficient

resources allocated by management also act as an obstacle for lean implementation (Aij et al., 2013; Mutingi et al., 2017; Ruiz & Ortiz, 2016; Stanton et al., 2014). According to Stanton, politics in the hierarchical organisational has a large number of dominant stakeholders which directly influence the availability of resources (Stanton et al., 2014). Although lean processes are perceived as low-cost interventions, an annual budget was suggested by respondents to be necessary for development of improvement activities and staff training (Ruiz & Ortiz, 2016). Lastly, sustenance of lean processes is another problematic issue for management and a challenging area which acts as a barrier to success of lean (Costa et al., 2015; DeSouza & Pidd, 2011; Fine et al., 2009; Kinder & Burgoyne, 2013).

To improve management practices for Lean implementation in healthcare requires strong leadership focusing on employee involvement, continuous improvement and employee empowerment. So, empowering frontline care staff to identify and address problems is essential for successfulness of lean implementation in healthcare. As research shows that lean management flourishes in the on bottom-up improvement in healthcare, management practices are needed to be grounded in organisational cultures where clinicians are given autonomy to participate in decisions on lean initiatives and implementation. Furthermore, as the lean implementation in healthcare often involves many stakeholders with differing interests and priorities, management practices need to value this diversity and manage these stakeholders effectively to achieve potential benefits of lean implementations (see Aij et al., 2013; Antonsen & Bye, 2020; Rosa et al., 2021; van Beers et al., 2022).

Organisational Structure. Organisation structure was found to be another important obstacle to successful lean implementation. The current work structure in healthcare is fragmented into different professional and functional groups giving rise to functional and professional silo imposing barriers to lean initiatives. Eight studies support that organisational silos are responsible for failure of lean processes (Aij et al., 2013; Costa et al., 2015; DeSouza & Pidd, 2011; Escuder et al., 2018; Halling & Wijk, 2013; Kinder & Burgoyne, 2013; Leite et al., 2024; Saha et al., 2024). Functional silos lead to fragmented care, which does not result into complete and total improvement of care provided to patient. Five studies highlighted that, different ways of working without inter-departmental collaboration and limited agreement between different professional groups also leads to sub-optimal improvements and failure of the whole lean initiative (Aij et al., 2013; Cheng et al., 2015; Grove et al., 2010; Halling & Wijk, 2013; Ruiz & Ortiz, 2016). Over this one leader has commented, *‘Sometimes you*

experience problems outside your circle of influence, and then you are stuck with a problem because you have not established an infrastructure that reflects collaborative work with other departments’ (Aij et al., 2013). In addition to above factors the complexity of work structure in healthcare organisations makes lean implementation more challenging (Costa et al., 2015; Edwards et al., 2011; Grove et al., 2010; Hastle et al., 2016). Unlike the assembly line of a manufacturing unit, healthcare organisations need to tailor their services based on patient’s needs and requirements. Both, the nature of work and different rationalities of different professional groups in healthcare doesn’t allow absorption of lean principles in some areas of healthcare organisation (Edwards et al., 2011) thus acting as an indirect barrier for lean management practices in healthcare setting. In the end, outsourcing in hospitals was also found to make lean implementation more difficult (Costa et al., 2015; Grove et al., 2010).

To improve the fragmented structures discussed above, healthcare organisations need to move to more value stream focussed approach. This value stream approach concentrations on patient journey from admission to discharge instead of focusing on individual professions and departments in healthcare organisations. At this point a patient is viewed as a primary customer and design workflows to meet patient’s needs and preferences. So, in this context healthcare professionals and departments can manage and improve value streams by fostering an atmosphere of cross-disciplinary and cross departmental collaboration. Such approach breaks down silos and creates opportunities for teams in professional groups and departments to make decisions and implement changes within their respective value streams, rather than merely relying on the instructions from top (Eriksson et al., 2016; Holden et al., 2015; Kellner et al., 2024; Lisiecka-Biełanowicz & Biechowska, 2025).

Employee Resistance and Poor Collaboration. Twelve studies found employee resistance to act as barrier to lean management (Aij et al., 2013; Costa et al., 2015; DeSouza & Pidd, 2011; Escuder et al., 2018; Fine et al., 2009; Fournier et al., 2021; Hastle et al., 2016; D. Hung et al., 2015, 2017; Jorma et al., 2016; Leite et al., 2024; Mutingi et al., 2017). Among all the professional groups, doctors and nurses have been reported to be most frequently against lean initiatives or projects (Jorma et al., 2016). A possible explanation for this is that lean has been perceived to be detached from clinical values and practices (Hastle et al., 2016). Besides, some physicians viewed lean initiatives as a potential threat to their authority. Their concern can be understood by the following quote from a physician, *‘I felt very much like, wait a minute here. I’m already being very efficient. I’ve been patted on*

the back for being efficient and having good patient satisfaction [scores], and now someone else who's not a physician is coming in to tell me how to do my work' (D. Hung et al., 2017). Other reasons for staff resistance have been described as, increase in documentation of work and reporting, fatigue over development projects and the will to stay in comfort zone and continue old ways of doing things (Hihnala et al., 2018; Jorma et al., 2016). Three studies also indicate that poor staff involvement and motivation also affect the outcome of lean initiatives. A top-down approach in lean initiatives also explains much of the resistance and decreasing level of workforce involvement in lean process (Hihnala et al., 2018; D. Hung et al., 2017; Kahm & Ingelsson, 2019). Furthermore, healthcare leaders have professional skills but lack motivational skills to coach and engage other employees which negatively affects the success of lean projects (Escuder et al., 2018).

The first step to overcome resistance to lean management is to understand the reasons for resistance. There may be many reasons. For example, clinicians may perceive that lean would target their medical practices or their decision-making autonomy, but they might not articulate these hidden fears and anxieties openly. So, understanding these hidden reasons is crucial for offsetting resistance. As a result, having a dialogue which enable all stakeholders to create collective insights and a common understanding of issues and opportunities in lean is important at this juncture. Schein (1993) argues that the process of dialogue creates a psychologically safe space for all the stakeholders share ideas and knowledge to build shared understanding of issues in organisations and foster creative thought. As a results, this process enables stakeholders from divergent backgrounds with conflicting interests to have a shared understanding about organisational change driven by lean management in healthcare. The end results of such understanding is reduction of employee resistance towards lean management in health with increased collaboration. Transformational leadership is essential to dialogue in healthcare organisations as it enable leaders to challenge assumptions held by professional groups in healthcare through discussions where professional groups and other employees in healthcare organisations feel comfortable and safe expressing their ideas, feelings and criticisms about lean management without fear of negative repercussions (see Fournier et al., 2021).

Discussion

This qualitative systematic review aimed at exploring the possible organisational barriers in lean implementation within healthcare settings. A thorough analysis revealed

most significant organisational barriers which limit the successful lean implementations in healthcare.

Based on the results of this review, poor perception and knowledge of lean has been found to be the most commonly discussed barrier in most reviewed studies. Consistent with the finding of this review, Joe (2013) suggests that it is highly essential to educate the healthcare staff about lean principles and methodologies to realise and sustain the benefits achieved in lean implementation. Conversely, training the staff in healthcare is another challenge because very few people in healthcare organisations have well founded knowledge and experience of lean principles, tools and methods of lean production. What complicates things more is when, educators from manufacturing background use unknown terminologies and lack examples from healthcare, making it more difficult to understand ideas and principles of lean (Pokinska, 2010). Also, at the top management level, inadequate business education and supply chain management capabilities among hospital-based buyers have been suggested to be potential barriers. (Burnes, 2001 cited in Parkhi, 2019) The very basic principle of lean is to comprehend value from customer's standpoint but defining 'customer' in healthcare is another problem faced by healthcare employees. This is because 'customer' is not a straightforward term in healthcare (Radnor et al., 2012). This uncertain notion of patient as customer and dynamics between different stakeholders acts as an obstacle in lean implementation process (Young & McClean, 2008). Hence staff education and encouragement is needed to overcome this issue. The staff training should be designed for better understanding of lean principles and tools by allowing them to make use of lean tools under supervision and support (Al-Balushi et al., 2014). Similar to the findings of this review, poor perception of lean management techniques has been considered to be a major barrier in acceptance of lean in healthcare. According to Kim et al. (2006), the term lean is misunderstood and some employees think of lean methods, as a way of working themselves out of employment. Hence it is important that the phrase is explained properly in context and application. Another inhibiting factor on the cultural front is the applicability of lean manufacturing methods and concepts to health sector with claims such as, 'humans are not automobile parts'. This can be overcome by staff training through which, the employees gradually understand the core values of lean which aims to remove waste processes to provide benefits (Pokinska, 2010). Lastly, the lack of evidence about improvements and increased efficiency by lean management techniques creates further problems in lean implementation (Winch & Henderson, 2009). In order to overcome this issue, record keeping and documentation of positive lean outcomes is necessary. Other

strategy to promote lean initiatives, is to take support from a successful lean organisation to avoid common mistakes in the early phase of implementation. This will increase reliability on lean thinking due to staff having a successful model of implementation (Spagnol et al., 2013). In all, a poor knowledge about lean and the wrong perception towards lean thinking has been found to act as a profound barrier.

The management practices have been identified as the second theme for this systematic review. In line with this result, management practices have been categorised as a key barrier to lean management in healthcare as well as in other industries. The balance of power strongly affects lean implementation. If the members in organisation resisting lean hold power, then it becomes difficult to bring the desired improvement and change of culture (Perderson & Huniche, 2011). Since physicians are delegated more authority in healthcare organisations in comparison to other employees, managers need to encourage lean and quality improvement initiatives among physicians and also involve them in the implementation process for the success of the project. (Abdallah, 2014) Furthermore, leadership skills for a lean culture needs teamwork, collaboration and effective communication which traditionally are not emphasised in physician's professional training (Demecko & Martinovsky, 2016). Full management support and commitment have been defined as key success factors which can enable staff to achieve and then maintain sustainable results. Hence, leadership can be considered as the missing link between academic lean management and an actual transformation of an organisation towards lean principles and practices (Rossum et al., 2016). The lack of support and management resistance towards lean processes has been found to exist in all functions of institute. This is primarily because of lack of clarity, insecurity of organisational change or clashing interests and also the challenge of learning something new. Therefore, it is essential for the healthcare managers to understand and give enough support to sustain lean thinking. (Jadhav et al., 2014) Another disabling factor discussed under this theme was the lack of resources such as time and funding for lean improvement initiatives. (Aij et al., 2013; D. Hung et al., 2015) In line with findings of this review (Jadhav et al., 2014; Perderson & Huniche, 2011), support and suggest that cost and time invested in lean programs are also responsible for failure in achieving lean healthcare. It is therefore necessary for hospital managers to reach a common understanding to solve problems and eliminate waste. The lean program should be connected to the hospital's improvement strategy to optimise the lean program. A clear strategic deployment plan with objectives will create a challenging vision and motivate the staff to achieve it, thus acting as an enabler to the overall lean

initiative (Noori, 2015). Lastly, similar to this study result (Hallam & Contreras, 2018) suggest there is limited evidence on sustenance and lasting effect of lean processes in most organisations. For overcoming this barrier, suitable leadership style and workforce flexibility are needed for transition from merely using technical lean tools to a transformed hospital culture of lean and continuous improvement (Rossum et al., 2016).

The next theme in discussion is organisational structure and the way it resists a successful lean implementation. Silos, both professional and functional in most healthcare organisations are responsible for compartmentalisation of improvement in processes which acts as a barrier for lean processes (Parkhi, 2019). Achieving success of lean requires hospital wide patient flow and improvement in outcome and care experience of patient. Hence, appreciation of the entire system of care is required because hospitals are interconnected and inter-dependent systems. Improvement in one section affects the operations and improvements in other departments as well (Patricia et al., 2017). Teamwork, communication and Joint problem solving are necessary for multi-disciplinary teams for achieving local application of lean (Andersen et al., 2013). A behavioural change and new ways of thinking is must for those who are used to narrowly think only about performance of their unit. Organisation as a whole has to be recognised and efforts are needed to break the silo mentality for success of lean implementation (Kim et al., 2006). Moreover, hospitals start implementing lean at department level but it needs to be followed sequentially through the entire system until the whole organisation has turned lean. Nevertheless, this is not easy in healthcare setting due to its highly political and complex structure composed of powerful professional groups and regulatory systems (Dominic, 2013). Therefore, it is clear that complexity of healthcare organisations complicates the application of lean. It has thus been suggested that lean improvement efforts should be altered to suit the complexity of care process and consider specific characteristics of the care delivery process before designing, implementing and evaluating lean processes (Mazzocato et al., 2014).

Finally, the last developed theme for this systematic review is, employee resistance and poor collaboration. Many studies suggest that employees: especially physicians and nurses are highly resistant to lean management (Aij et al., 2013; Jorma et al., 2016; Mutingi et al., 2017). In fact, resistance to any change is a common problem in healthcare organisations, and therefore many improvement projects fail due to resistant culture. Reason for resistance has been mainly documented as, the will to continue doing things in old ways, lack of knowledge about benefits of lean and difference of opinion in professional groups (Jorma et al., 2016). Another cause

described is the repetitive trials of new methods, programs and initiatives which ultimately makes the staff cynical and weary of new ideas. The staff perceive lean as the 'most recent idea' that will go away in years to follow, when executives will have something new again (Fillingham, 2007). Just like any other organisational change, the success of lean initiatives is highly influenced by the frontline staff's perception and their acceptance. Moreover, many studies claim that lean leads to work intensification and decreases worker control, affecting employee perception about lean (Holden et al., 2015). Also, staff's disbelief about the application of lean in healthcare due to its manufacturing background is a frequently documented barrier to lean. What further complicates this is that, lean in healthcare is often supported by industry consultants with limited familiarity of healthcare context, who use technical lean language. This creates negative reactions from healthcare employees (Drotz & Pokinska, 2014). Doctors and nurses are most resistant towards acceptance of lean culture since somebody from lesser educational and professional designation suggests the lean change initiative, and thereby threatens their authority. (Al-Balushi et al., 2014) The structure of healthcare organisations is hierarchical where physicians possess higher power and rank, whereas lean initiatives require collaboration, teamwork and an open communication. Additionally, inter-department communication is poor in healthcare due to complex and multiple units acting as a barrier to functional smoothness of lean processes (Pokinska, 2010). For lean to work in healthcare, a transformational leadership is needed. The first and foremost barrier necessary to overcome, is to convince employees that lean works in healthcare setting (Holden et al., 2015). This can be done by giving organisational examples which have been successful at lean initiatives. Additionally, encouraging employee's participation through training and acknowledging their success throughout the initial stages of lean application will increase their motivation (Al-Balushi et al., 2014). A shift is required from autocratic leadership to new leadership competencies that include teaching, mentoring and a leader who clearly expresses his experience and vision throughout organisation to facilitate formation of lean healthcare. A common vision that inspires all the employees and encourages them for desired behaviour can strengthen the capacity for change in the organisation (Rossum et al., 2016). Furthermore, effective communication is necessary at all levels and is a key success factor for lean projects. Good communication will facilitate employee involvement and cooperation with a shared understanding of organisational goal of lean improvement program (Noori, 2015). It has been suggested by many respondents that fostering a culture of innovation and collaboration is the key to overcome

most obstacles and facilitate successful transition to a lean healthcare system. (D. Hung et al., 2015).

Based on the result and discussion, the answer to research question has been obtained. It can thus be stated that major organisational barriers to lean implementation are: (1) Poor perception and knowledge of lean, (2) Management practices, (3) Organisational structure, (4) Employee resistance and poor collaboration.

This qualitative systematic review aimed at exploring the possible organisational barriers in lean implementation within healthcare settings and have found the following four factors: (1) Poor perception and knowledge of lean, (2) Management practices, (3) Organisational structure, (4) Employee non-co-operation and resistance.

Conclusion

This systematic review aimed at understanding the potential organisational barriers to the implementation of lean practices in healthcare. Lean management has been successful in other industries for more than 50 years, but the application of lean in healthcare domain is still novice with limited empirical research on its validity and effectiveness in healthcare. To maximise the potential for success of lean, it is highly essential to anticipate and realise the challenges to lean initiatives in healthcare settings.

Multiple factors accounted for the challenging application of lean in healthcare. Nevertheless, these challenges can be overcome by a system wide approach in staff education and training to give a clear understanding of lean principles and its benefits to improve employee perception and acceptance of lean. Altering the lean language for the context of healthcare will allow better permeability into healthcare organisations. Furthermore, physicians and other clinicians are very important stakeholders, therefore healthcare organisations need to encourage their involvement and leadership in lean initiatives and implementation for improved outcomes. Breaking silos by interdepartmental collaboration will facilitate communication and improve efficacy of lean processes throughout organisation. All the above, along with, careful 'adaptation' of lean tools and methods to the healthcare context will improve successful implementation of lean in healthcare.

The study nonetheless, has some limitations; the study population in the selected study papers mostly included top level executives, managers and physicians who are most influential in the organisation. Many frontline staff was not interviewed or surveyed in these study papers, which leaves out their opinion about what makes lean challenging for them. As a result, further study should be carried to understand these barriers from general employee perspectives, giving us insights about the real and root cause of these challenges.

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